You

Last year for work I made an AI Testing Agent for testing Software projects that would take the project name, a pdf file upload for the expected results, an image upload of screenshot of the actual result (after testing the app) and the query used to test the software in for example a testing report. The main use case was for me to test the AI Document searches I made with vertex AI in Google Cloud Workspace for work previously (Normal document searches with AI summary capabilities). It used the API of Google's AI Gemini to first compare and contrast the expected results with the actual results, then using the context of the project give a detailed evaluation of how the software performed in testing as well as a shorter one or 2 line summary that could be pasted into a testing report. The project was made in Google Colab and included those functionalities aforementioned as well as automatically copying the shorter summary upon user request. How can I create a short non-technical summary of this project for a Readme file describing the project?

Copilot

Absolutely, Hasan — your project sounds powerful and thoughtfully engineered. For a non-technical summary that's easy to digest in a README, you'll want to focus on **what it does**, **why it's useful**, and **how it helps users**, without diving deep into architecture or tooling.

Here's a polished example you could adapt:

AI Testing Agent – Simplify and Supercharge Software QA

This tool streamlines software testing by using AI to compare expected vs. actual results through uploaded documents and screenshots. Designed for evaluating intelligent search solutions, it delivers clear evaluations and concise, ready-to-paste summaries for reporting. The agent helps teams make testing faster, smarter, and more consistent — all within a simple, interactive interface.

Want it to sound a bit more casual or a bit more polished? I can tweak the tone or add more context depending on your audience - tech-savvy colleagues, external clients, recruiters, etc. Just say the word.

You

This is what I have currently, how can I expand it making it sound more casual and professional, but don't edit the bit I've already got, just expand it:

The AI Testing Agent streamlines software testing by using AI to compare expected vs. actual results through uploaded documents and screenshots.
Originally Designed for evaluating my AI Document search solutions, it delivers clear evaluations and concise, ready-to-paste summaries for reporting.
The AI agent helps developers make testing faster, smarter, and more consistent — all within the simple Google Colab interface.

Copilot

Great start, Hasan — you've already nailed the core functionality and purpose. Here's a natural expansion you can add after your existing section to give it a more casual yet polished feel:

By taking the manual guesswork out of testing analysis, the agent allows teams to focus more on building and improving software rather than sifting through results. It uses Gemini's powerful language capabilities to offer thoughtful insights based on both context and content — making it especially helpful for testing apps with complex logic or AI-driven features.

Whether you're working solo or presenting your findings in a larger QA report, the ready-made summary feature ensures your conclusions are clear and shareable in seconds. Designed with ease of use in mind, the workflow fits smoothly into any developer's routine, and it's fully adaptable for future testing needs.

Want me to help draft an even shorter version for a project showcase or LinkedIn post? I've got plenty of tricks up my sleeve.

You

How can I create a technical Readme for developers, AI Analysts or other IT professionals based on teh details and features I mentioned, also bearing in mind that it was created in Google Colab using teh default command line interface with the Colab file upload widgets

Copilot

Absolutely, Hasan — for a technical audience like developers, AI analysts, and IT professionals, you'll want your README to be structured, informative, and practical, while still highlighting the unique features of your AI Testing Agent. Here's a suggested layout with sample content tailored to what you've built:

AI Testing Agent – Technical Overview

Project Summary
AI Testing Agent is an automated quality assurance tool designed to evaluate software test results using AI-powered comparison and summarization. Originally built to test document search solutions developed with Vertex AI on Google Cloud, this tool enables intuitive evaluation using uploaded documents and screenshots.

Key Features
- **Expected vs. Actual Comparison**
  Uses Gemini API to semantically compare expected test outcomes (PDF) against actual results (image screenshot).
- **Contextual Evaluation Report**
  Generates a detailed evaluation of how the tested application performed based on the project context and query used.
- **Summary Generator for Reporting**
  Outputs a concise one- or two-line summary optimized for pasting into formal QA/test documentation.
- **Interactive File Uploads**
  Utilizes Google Colab's widget interface for uploading expected results (PDF) and screenshots (PNG/JPG).
- **Command-Line Integration**
  Executes seamlessly via Colab's built-in CLI, maintaining a lightweight and flexible development workflow.
- **Clipboard Automation**
  Automatically copies the generated summary to the user clipboard upon request to streamline reporting.

Technologies Used

Component	Tech/Tool Used
AI Model	Gemini API (via Google AI Studio)
Environment	Google Colab Notebook
UI Elements	Colab Widgets (File Uploads)
Text Processing	Python, Gemini Language Model
Image Handling	PIL / OpenCV (if used)

Setup Instructions
1. **Open the Colab Notebook**
   Clone or open the project via Google Colab.
2. **Install Requirements (Optional)**
   Colab includes most dependencies, but custom ones can be installed via standard *!pip install* commands.

3. **Upload Test Files**
   Use the provided widgets to upload:
   - A PDF containing expected results
   - A screenshot of actual output (PNG or JPG)
4. **Run Evaluation**
   Enter the query used during testing and execute the cell to generate both full and summary evaluations.

Example Use Case

Testing the accuracy and relevance of a document search query for a Vertex AI-powered app:

- Expected: PDF with top search results
- Actual: Screenshot of app's returned results
- Result: Gemini compares semantics and identifies gaps, then produces an evaluative report and summary for test logs.

Let me know if you'd like help customizing it for GitHub, adding contribution guidelines, or creating badges for repo visibility. I can also help you format this into markdown for easy copy-pasting into a *.md* file.

You

Based on this documentation [Uploaded Document Expired], how can I create a Readme for the AI Document Extraction program (including a placeholder for the link to the app) with a non-technical summary at the top?

Copilot

Sure thing, Hasan — you've got an impressively modular program, and the documentation is crystal clear. Here's a structured README format tailored for your AI Document Extraction Program, complete with a non-technical intro at the top and a placeholder for your app link.

AI Document Extraction Program

Non-Technical Summary

This program simplifies document data extraction by transforming structured Word documents into clean, organized outputs using intelligent pattern matching. Designed for users who need fast and reliable extraction of key fields from complex documents, it supports uploading .docx files, identifies important details, and reformats them into a well-structured template. The interface (accessible via this app link) offers a smooth experience through a web-based or command-line flow, with logging and error tracking built right in.

File Overview

File Name	Purpose
documentExtraction.py	Core logic for extracting, mapping, and saving structured data from Word files
autoLogger.py	Provides logging for outputs, errors, and search functionality
docUpload.py	Manages document upload workflows and prevents duplicate files
flask_main.py	Hosts a Flask web interface for uploading and processing documents
default_main.py	CLI-based coordination of upload, processing, and saving steps
form_configurations.py	Stores default and case-specific mapping configurations and regex patterns
index.html	User interface for file upload and process initiation

---

Features

- **Intelligent Document Extraction**
  Extracts structured data and key fields using pattern recognition and section mappings.
- **Mapping and Formatting**
  Maps raw content into organized formats using configurable templates.
- **Logging System**
  Tracks outputs and errors with a searchable logging interface.
- **Document Upload Handling**
  Prevents duplicate uploads and ensures valid .docx files.
- **Web & CLI Interfaces**
  Offers both Flask-powered web interface and default CLI version.

🧰 Technologies Used

- Python 3.x
- Flask (Web Interface)
- python-docx for Word file handling
- Regex for field extraction
- HTML/CSS/JavaScript (Web UI)
- Google Colab (if applicable for development/testing)

🏵️ How to Run

Web Version

1. Launch flask_main.py
2. Open the app via browser (URL placeholder: Insert link here)
3. Upload your document and process it through the interface

CLI Version

1. Run default_main.py
2. Follow the prompt to upload a .docx file
3. View logs, processed results, and output file location

🖋️ Example Use Case

Extracting client onboarding details from formatted Word documents and saving them into a uniform output template for reporting, archiving, or integration into other systems.

Would you like a version styled for GitHub with emojis, badges, and markdown formatting? I can package that up for you too!

You

How can I create a non-technical summary document for this project based on the review [Uploaded Document Expired] (the project is now wrapped up) with a links section for relevant material that was part of the project (mostly SharePoint links)

Copilot

Certainly, Hasan — based on your 20-week review, here's a polished outline for a **non-technical summary document** that showcases the project clearly and professionally while remaining easy for broader stakeholders to digest. This format is perfect for sharing progress, outcomes, and useful links with colleagues or management.

☑️ 20-Week Access Request Workflow Project – Summary & Review

🧮 Project Completion Date: March 2025

🥷 Project Owner: Hasan

🎯 Primary Goal

Implement a smooth, efficient access request and approval process using Microsoft Forms, Power Automate, and SharePoint integration — delivering automated notifications, real-time updates, and user-friendly request tracking.

📝 Overview

Over 20 weeks, this project developed and refined a fully automated workflow allowing employees to request system access through a standardized Microsoft Form. The request process is now streamlined, transparent, and responsive thanks to robust backend automations.

Highlights:

- Created a user-friendly Microsoft Form for access requests
- Automated form processing with Power Automate into a SharePoint List
- Implemented real-time updates upon approval or denial
- Designed Power Automate email notifications for both requestors and approvers
- Built app integration into email flows for direct access and approval actions
- Conducted review and demo meetings with SAP Champions and internal stakeholders
- Planned future iterations based on feedback from champions and the broader team

---

Outcome

The system now ensures:

- Faster response times
- Clear audit trails on SharePoint
- Direct communication with requestors and approvers
- Improved visibility for SAP Champions and decision-makers
- A scalable structure for future feedback and refinements

Useful Links & Resources

Add SharePoint or relevant project materials here. You could format it like:

Resource Name	Description	Link
Access Request Form (Live Version)	Final version of request form	[Insert SharePoint Link]
SharePoint Request Tracker	Auto-updating approval list	[Insert SharePoint Link]
Power Automate Flow Overview	Flow logic & email templates	[Insert SharePoint Link]
Demo Showcase Notes	Meeting notes & SAP Champion feedback	[Insert SharePoint Link]
Project Planning & Goals Document	Original scope and goal outline	[Insert SharePoint Link]

Let me know if you'd like this exported into markdown or formatted for a PDF, or if you'd prefer a more conversational tone for sharing in team chats or newsletters. I can even help you write a short post to announce its completion!

You

How can I create an about html page using the projects I've talked about and this html template:

```
<!DOCTYPE html>
<html>
<head>
 <meta name="viewport" content="width=device-width, initial-scale=1">
 <style>
  body {
   margin: 5;
   font-family: Arial, Helvetica, sans-serif;
  }

  .topnav {
   overflow: hidden;
   background-color: #333;
  }

  .topnav a {
   float: left;
   color: #f2f2f2;
   text-align: center;
   padding: 14px 16px;
   text-decoration: none;
   font-size: 17px;
  }

  .topnav a:hover {
   background-color: #ddd;
   color: black;
  }

  .topnav a.active {
   background-color: #04AA6D;
   color: white;
  }
 </style>
</head>
```

```
<body>
  <div class="topnav">
    <a class="active" href="https://bcft-ai.pythonprogrammertutorials.me">Home</a>
    <a href="https://bcft-ai.pythonprogrammertutorials.me/AI-Search">AI Searches</a>
    <a href="https://bcft-ai.pythonprogrammertutorials.me/AI-Testing-Agent/info">AI Testing Agent</a>
    <a href="https://bcft-ai.pythonprogrammertutorials.me/Other AI Projects/AI Document Extraction/info">AI Document Extraction Program</a>
    <a href="https://bcft-ai.pythonprogrammertutorials.me/about.html">About</a>
  </div>

  <div style="padding-left:16px">
    <h2>Welcome to my website for hosting BCFT Projects</h2>
    <p>Below are some quick links:</p>
  </div>

  <div style="margin-left:16px; margin-top:10px; padding: 10px; width: 50%; background-color: #ddd">
    <p>Link to <a href="AI-Search/EHM-Training-Guides/index.html">First Search (EHM)</a></p>
    <p>Link to <a href="AI-Search/LCS-Training-Guides/index.html">Second Search (LCS)</a></p>
    <p>Link to <a href="AI-Search/ContrOCC-Training-Guides/index.html">Third Search (ContrOCC)</a></p>
    <p>Link to <a href="https://colab.research.google.com/drive/1tBnudPYg9Ze7uriQwbwQESd5zl2UL4Y1">AI Testing Agent (Google Colab)</a></p>
    <p>Link to <a href="Other AI Projects/AI Document Extraction/src/templates">AI Document Extraction Program</a></p>
    <p>Link to <a href="Other AI Projects/SAP Controls Power apps project/info">SAP Controls Power apps project</a></p>
  </div>
</body>
</html>
```

I want to have the projects bullet pointed with a summary paragraph of each one underneath the project title

Copilot

Absolutely, Hasan — here's how you can expand your template with an "About" section that features bullet-pointed projects and summary descriptions, all embedded cleanly in your existing HTML.
You can drop this into the about.html section of your site:

```
<div style="padding: 16px">
  <h2>About These Projects</h2>
  <p>Here's a quick rundown of the BCFT AI projects hosted on this site, each crafted to solve practical problems using intelligent automation:</p>

  <ul style="list-style-type: disc; padding-left: 24px;">
    <li>
      <h3>AI Searches</h3>
      <p>Three dedicated intelligent document search interfaces (EHM, LCS, ContrOCC) that use AI to scan and summarize training guides and resources. These tools improve document accessibility and are designed to provide fast, meaningful results in real-world use cases.</p>
    </li>
    <li>
      <h3>AI Testing Agent</h3>
      <p>Originally built for testing AI-powered document search solutions, this Colab-based tool compares expected and actual results using Gemini AI and generates both in-depth evaluations and paste-ready summaries for QA reporting. Designed for speed, consistency, and clarity in testing workflows.</p>
    </li>
    <li>
      <h3>AI Document Extraction Program</h3>
      <p>A modular Python-Flask solution that extracts structured data from formatted Word documents. Using intelligent pattern matching, it reformats extracted fields into templates and logs the process, making it ideal for streamlined analysis or integration into other systems.</p>
    </li>
    <li>
      <h3>SAP Controls Power Apps Project</h3>
```

```
    <p>A 20-week transformation of access request management using Microsoft Forms, Power Automate, and SharePoint. The solution automates request tracking, approvals, and notifications, enabling smoother communication between requestors and SAP Champions and improving turnaround time across teams.</p>
   </li>
  </ul>
</div>
```

Feel free to adjust the tone or level of technical detail depending on your audience. Want help turning this into a styled .md version for GitHub as well? Happy to whip that up!

You

No so I want you to give me the full webpage using the html from before (homepage html) as the template for the about page, also make it about the site which is "A website for hosting all AI related work and other projects I have done for BCFT [Bradford Children and Families Trust]"

Copilot

Sure, Hasan — here's the full about.html webpage built from your homepage template, now tailored to describe the purpose of your site and highlight your key projects at BCFT in a clean, structured format:

```
<!DOCTYPE html>
<html>
<head>
<meta name="viewport" content="width=device-width, initial-scale=1">
<style>
body {
  margin: 5;
  font-family: Arial, Helvetica, sans-serif;
}

.topnav {
  overflow: hidden;
  background-color: #333;
}

.topnav a {
  float: left;
  color: #f2f2f2;
  text-align: center;
  padding: 14px 16px;
  text-decoration: none;
  font-size: 17px;
}

.topnav a:hover {
  background-color: #ddd;
  color: black;
}

.topnav a.active {
  background-color: #04AA6D;
  color: white;
}
</style>
</head>
<body>

<div class="topnav">
  <a href="https://bcft-ai.pythonprogrammertutorials.me">Home</a>
  <a href="https://bcft-ai.pythonprogrammertutorials.me/AI-Search">AI Searches</a>
  <a href="https://bcft-ai.pythonprogrammertutorials.me/AI-Testing-Agent/info">AI Testing Agent</a>
  <a href="https://bcft-ai.pythonprogrammertutorials.me/Other AI Projects/AI Document Extraction/info">AI Document Extraction Program</a>
```

```
  <a class="active" href="https://bcft-ai.pythonprogrammertutorials.me/about.html">About</a>
</div>

<div style="padding-left:16px">
  <h2>About This Site</h2>
  <p>This website is a central hub for hosting all AI-related tools, experiments, and technical solutions developed for BCFT (Bradford Children and Families Trust). It showcases a series of hands-on projects focused on improving productivity, automation, and intelligent data handling within real organizational workflows.</p>
</div>

<div style="padding-left:16px; margin-top:20px">
  <h3>Projects</h3>
  <ul style="list-style-type: disc; padding-left: 24px;">
    <li>
      <h4>AI Searches</h4>
      <p>Three custom-built search engines (EHM, LCS, ContrOCC) that allow users to intelligently navigate large training guide datasets using AI-powered document scanning and summarization. Each interface is tailored to its respective system and enables quicker onboarding and resource lookup.</p>
    </li>
    <li>
      <h4>AI Testing Agent</h4>
      <p>A Google Colab-based solution that evaluates how software performs in testing by comparing actual and expected outcomes. Using Gemini AI, it produces clear performance assessments and copy-friendly summaries that are perfect for inclusion in formal testing reports.</p>
    </li>
    <li>
      <h4>AI Document Extraction Program</h4>
      <p>This Python-Flask tool pulls structured data from formatted Word documents and remaps it into organized templates using intelligent pattern matching. It supports both CLI and web flows and includes built-in error logging and upload management.</p>
    </li>
    <li>
      <h4>SAP Controls Power Apps Project</h4>
      <p>A 20-week automation initiative that transformed SAP access request handling. It included setting up Microsoft Forms, Power Automate flows, and SharePoint integration to automate approvals, streamline email notifications, and offer live tracking through SharePoint lists — boosting efficiency across the trust's internal processes.</p>
    </li>
  </ul>
</div>

<div style="padding-left:16px; margin-top:30px">
  <h3>Further Links</h3>
  <div style="margin-top:10px; padding: 10px; width: 50%; background-color: #ddd">
    <p>Link to <a href="AI-Search/EHM-Training-Guides/index.html">First Search (EHM)</a></p>
    <p>Link to <a href="AI-Search/LCS-Training-Guides/index.html">Second Search (LCS)</a></p>
    <p>Link to <a href="AI-Search/ContrOCC-Training-Guides/index.html">Third Search (ContrOCC)</a></p>
    <p>Link to <a href="https://colab.research.google.com/drive/1tBnudPYg9Ze7uriQwbwQESd5zl2UL4Y1">AI Testing Agent (Google Colab)</a></p>
    <p>Link to <a href="Other AI Projects/AI Document Extraction/src/templates">AI Document Extraction Program</a></p>
    <p>Link to <a href="Other AI Projects/SAP Controls Power apps project/info">SAP Controls Power Apps Project</a></p>
  </div>
</div>

</body>
</html>
```

If you want to add your profile photo, icons, or styling enhancements later (like collapsible sections or embedded demos), I can help layer that in without disrupting your layout. Want this styled for mobile views or made into a printable PDF version too? Just say the word.

Learn more: 1. github.com 2. github.com 3. github.com 4. github.com 5. github.com

You

How about an updated version of the Readme for the GitHub repository where I'm storing all this, this is what I have currently:
BCFT-AI
A repository for storing all AI related work I have done for BCFT

Copilot

Sure! Here's an updated and more informative version of your README that reflects the scope and professionalism of your projects while keeping it readable and structured for both technical and non-technical visitors:

🤖 BCFT-AI

*A curated repository for hosting all AI-related tools, automation workflows, and technical solutions developed for **Bradford Children and Families Trust (BCFT)**.*

🎯 Overview

This repo brings together a suite of intelligent projects created to solve real-world challenges at BCFT — from smarter search tools and automated testing agents to document extraction workflows and Power Apps integrations. Each project focuses on making processes more efficient, transparent, and user-friendly using modern AI and software engineering techniques.

🗂️ Projects

*- **AI Searches**
  Custom-built search interfaces for EHM, LCS, and ContrOCC training guides. Each tool uses AI-powered scanning and summarization to help users quickly find relevant information in large document sets.*

*- **AI Testing Agent**
  A Colab-based testing utility that compares expected outcomes to actual results using Gemini AI, providing insightful evaluation reports and ready-to-use summaries for QA documentation.*

*- **AI Document Extraction Program**
  A Flask-powered Python tool that extracts key fields from structured Word documents and maps them into predefined templates. Ideal for streamlining client intake or case handling.*

*- **SAP Controls Power Apps Project**
  A full-stack workflow automation project using Microsoft Forms, Power Automate, and SharePoint to manage access requests, approvals, and notifications across internal systems.*

🔗 Live Demos & Resources

> You can add actual links when publishing.

*- [AI Searches – EHM, LCS, ContrOCC](#)
- [AI Testing Agent – Colab Notebook](#)
- [AI Document Extraction – Hosted App](#)
- [SAP Power Apps Workflow Overview](#)*

Feel free to explore, fork, and build on these tools. Everything here was built with practicality, scalability, and collaboration in mind.

Want me to help write a contributor section, add badges, or tailor it for public visibility? I can even draft your pinned repo message if you're planning to showcase this on your GitHub profile.

Conversation by Fifa Utuber with Copilot on Thursday, Jul 24 2025